

their ADAS solutions.

Autonomous driving. IMU operates independently of external conditions, relying on gravity and the laws of physics; hence in the event of other sensors' failure or worst weather conditions, IMU continues to provide data ensuring safety until the vehicle comes to a secure stop or the complete system resumes functioning. By eliminating data interruption and increasing operational safety, the IMU accelerates the realisation of Level 5 fully autonomous driving. The accuracy of an IMU's performance is solely determined by the precision of its own sensor's acceleration measurements. The IMU is considered more affordable compared to other sensors in ADAS. A high-performance IMU plays a vital role in achieving highly accurate positioning in autonomous vehicles.

The way ahead

IMU-equipped ADAS holds the potential for development and advancement in future. Nowadays, engineers work on designing the automatic communication between cars, called Vehicle to Vehicle (V2V), and between cars and surrounding infrastructure, called Vehicle to Infrastructure (V2I). These advancements in automotive technology are revolutionising how vehicles interact with each other and their environment, paving the way for safer and

Challenges and opportunities

The automotive industry is undergoing a significant transformation with the introduction of electric vehicles (EVs), bringing challenges and opportunities for the use of IMUs in advanced ADAS systems.

Challenges

- The complex electrical/electronic architectures in vehicles include many electronic control units (ECUs) hindering innovation.
- A paradigm shift is needed from distributed hardware to software and services that pave the way for new opportunities in ADAS development within electric vehicles.
- Ensuring the accuracy and calibration of IMU sensors in varying environmental conditions is a different challenge.
- Integrating IMU data with inputs from other sensors requires sophisticated sensor fusion algorithms and precise calibration.
- Processing the vast amounts of real-time data generated by IMUs demands efficient algorithms and computational resources.

Opportunities

- IMU-equipped ADAS systems can enhance safety by accurately measuring a vehicle's motion, enabling collision avoidance, lane-keeping assist, and emergency braking.
- Vehicle dynamics and control can be improved, leading to optimised traction, stability, and adaptive suspension systems.
- IMUs are crucial in autonomous driving, enabling precise localisation, mapping, and path planning.
- Sensor fusion with IMU data enhances the perception and object detection capabilities.

more efficient driving experiences. However, IMUs are not directly involved in the communication aspect. They provide crucial data supporting accurate positioning, orientation, and motion tracking, which is essential for effective V2V and V2I interactions. The development of advanced motion control algorithms can utilise IMU data to optimise vehicle stability. Integrating artificial intelligence (AI) and machine learning (ML) techniques with IMU data will enable intelligent and adaptive ADAS functionalities. Finally, the ongoing development

includes evolving safety features, such as collision detection and avoidance systems, pedestrian and cyclist detection, and improved emergency braking systems.

IMUs have already made a significant impact on ADAS. Looking to the future, IMUs can further enhance ADAS systems in several ways. IMUs are expected to benefit from advancements in sensor technology, including higher accuracy, improved reliability, and reduced power consumption. This will enable them to contribute to developing advanced safety features in ADAS. IMUs will remain integral to achieving high-precision positioning and navigation in autonomous driving. The miniaturisation of IMUs and advancements in manufacturing techniques will likely lead to smaller, more affordable, and mass-market-ready IMU solutions. This will facilitate their widespread adoption in various ADAS applications, making them a standard component in future vehicles. **EFY**



The author, Nidhi Agarwal, works as a tech journalist at EFY